# BMI Calculator Project

*README*

## Project Overview

The BMI Calculator is a Python console-based project that collects basic user information, calculates Body Mass Index (BMI), displays the result, assigns a BMI category using the thresholds programmed in the project, and identifies an age group. The program can be run repeatedly until the user chooses to stop.

## Purpose of the Project

The main purpose of this project is to demonstrate how Python can be used to create a simple interactive application. It brings together user input, data conversion, functions, arithmetic operations, conditional statements, loops, validation, and formatted output in one program.

## Main Features

- Collects name, age, gender, height and weight from the user.
- Converts age to an integer and height and weight to decimal values.
- Converts height from centimetres to metres before calculating BMI.
- Calculates BMI and rounds the result to two decimal places.
- Displays the user's entered information and calculated result.
- Uses conditional statements to assign the BMI category programmed in the code.
- Displays a BMI range reference.
- Identifies the programmed age group: Child, Teenager, Adult or Senior.
- Checks for non-positive height or weight values.
- Allows the user to perform another calculation.

# How the Program Works

1. The program displays a welcome message.

2. The **get_details()** function asks the user for personal and measurement details.

3. The program checks whether height and weight are greater than zero.

4. The **calculate_bmi()** function converts height to metres and calculates BMI.

5. The **show_result()** function presents the result and applies the programmed BMI and age-group conditions.

6. The user is asked whether another calculation is required.

7. The loop continues for “yes”, stops for “no”, and ends for any other response.

# Python Concepts Used

The project demonstrates functions, variables, input and output, type conversion, arithmetic operators, return values, if/elif/else statements, a while loop, string methods, comparison operators, validation, and formatted console output.

# Project Note

The BMI categories and suggestions in this project are the rules written into the program for educational demonstration. BMI is a screening measure and should not be treated as a complete assessment of a person's health, especially for children and teenagers.